Pokémon Collection

How to install and run the application on your own computer (Windows)

Pokémon Collection is a small web application that runs entirely on your computer: a local server reads and writes one database file, and you use it from your web browser. Nothing is sent to the internet while you use it. Installation takes about ten minutes.

What you will end up with

- The application in a folder of your choice on your computer. This guide uses `Documents\Dev_projects\Pokemon` inside your own user folder as the example.
- Your own data in one file: `data\pokemon_collection.db` inside that folder. Each person who installs the application has their own separate collection.
- The application in your browser at `http://127.0.0.1:8000` whenever the server is running

1. Before you start

You need two free tools. All commands in this guide are typed in **PowerShell** (Start menu → type "PowerShell" → Enter). Type or paste one command at a time and press Enter.

About folder names in this guide

`$HOME` is PowerShell's shortcut for your own user folder (for example `C:\Users\YourName`), so the commands work as written for any person on any computer — you do not need to replace it. If you prefer another location, replace `$HOME\Documents\Dev_projects` with your own path everywhere it appears, and keep the quotation marks.

1 Git — downloads the application and its updates

Check whether it is already installed:

```
git --version
```

If you see a version number, skip ahead. Otherwise install it, then **close and reopen PowerShell**:

```
winget install --id Git.Git -e
```

(Alternative: download the installer from `https://git-scm.com/download/win` and accept the default options.)

2 uv — installs Python and the application's components

Check whether it is already installed:

```
uv --version
```

If not, install it, then **close and reopen PowerShell**:

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

You do not need to install Python yourself: uv downloads a suitable version (3.12 or newer) automatically if your computer does not have one.

2. Install the application

3 Create the folder that will hold the application

Create a parent folder (nothing happens if it already exists):

```
New-Item -ItemType Directory -Force "$HOME\Documents\Dev_projects"
```

Point PowerShell at it. `cd` means “change directory”: every following command runs inside that folder.

```
cd "$HOME\Documents\Dev_projects"
```

The start of the PowerShell line now shows the folder you are in, for example `PS C:\Users\YourName\Documents\Dev_projects>`. You can also display it at any time with:

```
pwd
```

4 Download the application into it

```
git clone https://github.com/kzandstra/cards.git Pokemon
```

- This creates a new `Pokemon` folder inside `Dev_projects`. It must not already exist; if it does, rename the old one first or choose another name in place of `Pokemon`.
- If the repository is private, the owner must first give your GitHub account access to it, and a window asks you to sign in to GitHub the first time.

5 Go into the application folder and install the components

```
cd "$HOME\Documents\Dev_projects\Pokemon"
```

```
uv sync
```

This creates a `.venv` folder with everything the application needs. It only has to be done once, and again after an update.

3. Start and use the application

6 Start the server

Each time you open a new PowerShell window, first point it at the application folder, then start the server:

```
cd "$HOME\Documents\Dev_projects\Pokemon"
```

```
uv run uvicorn app.main:app --reload
```

Wait for this line to appear:

```
Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
```

7 Open the application

In Firefox (or any browser) open `http://127.0.0.1:8000`. The badge at the top right shows "Connected · 1789 cards".

> **Keep the PowerShell window open**
> The application only works while that window is running the server. If you close it, the browser shows "Unable to connect". To stop the server on purpose, click in the window and press **Ctrl + C**. To use the application again later, repeat step 6.

> **Shortcut: open PowerShell directly in the folder**
> Open the application folder in File Explorer, click in the address bar at the top, type `powershell` and press Enter. PowerShell opens already pointed at that folder, so you can skip the `cd` command.

First steps in the application

- **Add a card you own:** pick a set, find the card, press "+ I have this card", then set quantity and condition. It is saved immediately.
- **Card or set missing?** Use "➕ Add a card" or "➕ New set" next to the search box. Cards and sets you added can be corrected or deleted with their ✎ button.
- **Sell a card:** press "🏷 Sell" on a card you own, then follow it in the Sales tab.
- The "❓ How do I add a card?" panel at the top of the Collection tab explains each step.

4. Back up your data

Everything you enter lives in `data\pokemon_collection.db`. Make a copy regularly, and always before an update (run this in the application folder):

```
uv run python scripts/backup_db.py
```

The copy is written to the `backups` folder with the date and time in its name. You can also download a copy from the application: Export / Database tab → "SQLite database (.db)".

5. Update to a newer version

Stop the server (Ctrl + C), then:

```
cd "$HOME\Documents\Dev_projects\Pokemon"
```

```
uv run python scripts/backup_db.py
```

```
git pull
```

```
uv sync
```

```
uv run uvicorn app.main:app --reload
```

Then reload the page in the browser (Ctrl + F5).

If `git pull` refuses because of `data/pokemon_collection.db`

Your database contains your own cards, so Git sees it as modified. This only blocks the update when the database file was also changed in the new version. Do not delete anything. Put your database aside, update, then put it back:

```
copy data\pokemon_collection.db "$HOME\Documents\pokemon_collection_mine.db"
```

```
git checkout -- data/pokemon_collection.db
```

```
git pull
```

```
copy "$HOME\Documents\pokemon_collection_mine.db" data\pokemon_collection.db
```

The application adds any new tables to your database by itself the next time it starts.

6. If something goes wrong

What you see	What to do
Browser: “Unable to connect” to 127.0.0.1:8000	The server is not running. Do step 6 and wait for the “Uvicorn running” line. If the command stopped with an error, read the last lines in PowerShell and look them up below.
`git` or `uv` “is not recognized”	The tool is not installed, or PowerShell was opened before it was installed. Close PowerShell, reopen it, and try again; otherwise redo step 1 or 2.
“destination path already exists and is not an empty directory”	A `Pokemon` folder already exists there. Rename it, or use another name in the command, then repeat step 4.
“Cannot find path … because it does not exist” after `cd`	The folder name in the command does not match where you installed the application. Find the folder in File Explorer, then use its real path in the `cd` command (in quotation marks), or use the address-bar shortcut from section 3.
“address already in use” / port 8000 is busy	Another copy of the server is still running. Close the other PowerShell window, or start on another port by adding `--port 8001` to the command in step 6 and opening `http://127.0.0.1:8001`.

“Could not import module app.main” or “No module named app”	PowerShell is not pointed at the application folder. Run the `cd` command from step 6 first, and check with `pwd`.
Badge shows “Cannot reach the API” or “Save failed”	The server was stopped while the page was open. Restart it (step 6) and reload the page.
The page looks unchanged after an update	Reload with Ctrl + F5 so the browser fetches the new version.
A red error message (traceback) when starting	Copy the last 10–20 lines; they say exactly what failed. Your data is not affected by a failed start.

7. Command summary

To do this	Run this in the application folder
Start the application	`uv run uvicorn app.main:app --reload`
Stop the application	Ctrl + C in the PowerShell window
Back up the database	`uv run python scripts/backup_db.py`
Get the latest version	`git pull` then `uv sync`
Refresh market prices (needs internet)	`uv run python scripts/refresh_prices.py` — back up first
Run the automated tests	`uv run pytest -q`

Source code: https://github.com/kzandstra/cards · The application listens only on 127.0.0.1, so it is reachable from this computer only.